RONAN-ANDREW ALO

He/Him
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EDUCATION

Cornell University, Sibley School of Mechanical and Aerospace Engineering, *Ithaca, NY* *May 2028*
Candidate, Mechanical Engineering, Computer Science, 3.061 GPA

Marine Academy of Science and Technology, *Highlands, NJ* *June 2024*
AP Calculus BC, AP Physics C, Systems Engineering II, 94.10 Unweighted GPA

PROJECTS

'Plug and Scrub' *March 2023 – June 2024*

- Researched, designed, constructed, tested, and developed a car exhaust filtration device to protect drivers from CO poisoning/other pollutants during engine activation in closed spaces
- Finalized the design as a 3-inch diameter cylindrical housing that contained two filters in series, connected to a vehicle via a custom adapter and fitted with an exhaust fan on the opposite end
- Manufactured a model along with an initial and an improved working successor prototype
- Received 3rd in Community Sustainability Challenge Scholarship from Solar Landscape

Calculator in Java *March 2025*

- Coded a calculator in Java which can take inputs in RPN and standard notation
- Utilized class organization, stacks, and singly linked nodes

'Pacman' in Java *May 2025*

- Coded Pacman with randomized hills on maps which affect speed of characters
- Utilized a decreasing-key priority queue data structure, implemented a variant of Dijkstra's shortest path algorithm

SKILLS

- 4+ Years Experience Fusion 360
- 1 Year Experience Java, Python
- 2+ Years Experience Technical Writing

WORK EXPERIENCE

Cornell Recreational Services, *Cornell University* *September 2025- Present*
Fitness Center Monitor

LEADERSHIP EXPERIENCE

Athletics Team, *NJROTC* *May 2023 – May 2024*
Athletics Team Lead

- Devised new daily physical training schedule for 50+ cadets
- Personally trained with team 5 days a week before/within school hours and 2 days a week after school
- Organized school wide competitions and team participation in inter-unit competitions

Student Council
Vice President *April 2023 – June 2024*

- Designed and assisted in organizing fundraisers and events/ supporting the president in leadership duties, coordinating meetings, and promoting student involvement

Secretary *April 2022 – June 2023*

- Recorded minutes, directed bi-monthly meetings while managing correspondence, preparing agendas, supporting event planning/coordination within student council

EXTRACURRICULARS & VOLUNTEER SERVICE

SAT Math Section Tutor, *Schoolhouse* *Fall 2023*
Volunteer/Member, *Circle K* *Fall 2024-Present*